Cross-Framework Quantum Algorithm Benchmarking via Unified Compilation to OpenQASM 3.0

Umer Farooq[1], Hussan Waseem Syed[1], and Muhammad Irtaza Khan[1]

Department of Computer Science, FAST-NUCES, Islamabad, Pakistan
{umer.farooq, hussan.syed, irtaza.khan}@nu.edu.pk

Abstract. Cross-framework comparisons of quantum software development kits (SDKs) are methodologically compromised by a simulator confound: each framework is typically evaluated on its own native backend, making it impossible to attribute observed differences to the transpiler rather than the simulator. We resolve this by introducing a unified benchmarking pipeline built on QCanvas, which compiles circuits from any source framework to OpenQASM 3.0 and routes all execution through a single QSim statevector backend. We apply this pipeline to 12 quantum algorithms across 3 frameworks (Qiskit, Cirq, PennyLane), up to 8 qubits, at 4096 shots, and across 6 metric dimensions. Three findings stand out. First, Qiskit and Cirq are virtually equivalent in compiled circuit quality (QPack S_{overall} 58.80 vs. 58.79), differing only in ancilla-placement strategy on error-correction circuits. Second, PennyLane exhibits super-linear gate scaling on oracle and search algorithms: for Grover's algorithm the power-law exponent reaches a=2.21, yielding 216 gates at 6 qubits versus 18 for Qiskit and Cirq. Third, Jensen–Shannon divergence equals exactly 1.000 between PennyLane and both other frameworks for six algorithms at 4 or more qubits, indicating fully non-overlapping measurement distributions—a systematic decomposition incompatibility, not noise. NISQ practitioners should treat Qiskit and Cirq as interchangeable in circuit-quality terms and select PennyLane only when its differentiable programming model is required, accepting the overhead at scale.

Keywords: quantum benchmarking · framework comparison · OpenQASM 3.0 · NISQ · circuit transpilation

1 Introduction

The quantum software ecosystem has fragmented into a collection of competing, incompatible SDKs. Qiskit [8], Cirq [2], and PennyLane [1] are the three most widely adopted frameworks, each offering a distinct programming model and each producing a different gate decomposition for the same conceptual circuit. For a practitioner selecting a framework for a NISQ application, this diversity creates a genuine engineering problem: performance differences reported in the literature may reflect the choice of simulator as much as the choice of SDK.

Prior cross-framework comparisons are methodologically flawed in a consistent way: each framework is executed on its own native simulator. A difference in measured fidelity could originate from the transpiler, the simulator, or an interaction between the two. In the NISQ era—where gate counts and circuit depth directly govern decoherence and error accumulation [7]—this confound is not a minor caveat; it invalidates the comparison as a guide to framework selection.

We resolve the confound with QCanvas, a unified compilation platform that acts as a single intermediate layer between framework source code and simulation. The pipeline is: native circuit (Qiskit, Cirq, or PennyLane) → QCanvas converter → OpenQASM 3.0 [3] → QSim statevector backend. The backend is fixed for all frameworks; every measured difference is therefore attributable solely to the framework's transpilation strategy and gate vocabulary.

This paper makes four contributions:

1. A unified benchmarking methodology that eliminates the simulator confound via a common OpenQASM 3.0 intermediate representation.
2. Empirical evaluation of 12 quantum algorithms across 6 metric dimensions: compilation time, circuit structure, scaling behaviour, simulation performance, statistical equivalence, and holistic QPack scores.
3. Discovery of systematic decomposition incompatibilities in PennyLane that produce JSD= 1.000 divergence at scale, with documentation of which algorithms and qubit thresholds trigger them.
4. An open benchmark suite and automated notebook pipeline for reproducible framework comparisons.

Section 2 surveys related work. Section 3 provides background on the three frameworks, OpenQASM 3.0, and the metrics used. Section 4 describes the QCanvas pipeline and measurement protocol. Section 5 presents results across all six metric dimensions. Section 6 discusses implications for framework selection and identifies limitations. Section 7 concludes.

2 Related Work

2.1 Framework-Specific Benchmarking

A substantial body of work characterises the performance of a single quantum SDK in isolation. Studies of Qiskit's transpiler optimisation levels show that higher optimisation passes reduce CNOT counts by 10–40% on typical circuits, but these evaluations run exclusively on Qiskit's Aer simulator and cannot be compared directly to equivalent Cirq or PennyLane measurements. Similarly, PennyLane benchmarks in the variational literature report gradient-evaluation throughput on its `default.qubit` backend, a metric with no direct counterpart in Qiskit or Cirq. Framework-specific benchmarks are valuable for intra-framework optimisation studies; they do not establish cross-framework rankings.

2.2 Cross-Framework Comparisons

Multi-framework studies exist but carry the simulator confound described above. The most common experimental design pairs each framework with its native backend (Qiskit + Aer, Cirq + cirq-google simulator, PennyLane + default.qubit) and reports fidelity or runtime figures side by side. An observed difference in simulation time between Qiskit and Cirq in such a study conflates transpiler latency, compiled circuit size, and the throughput of two distinct simulators—three variables that cannot be disentangled post hoc. A second limitation of existing cross-framework studies is scope: most restrict comparison to one or two algorithms (typically Bell state and Quantum Fourier Transform) without systematically varying qubit count, which precludes scaling analysis. To our knowledge, this paper presents the first study that routes all circuits from all frameworks through a single common backend before comparison.

2.3 Quantum Volume and Holistic Benchmarks

Cross et al. introduced Quantum Volume (QV) as a single-number estimator of a device's effective circuit-processing capacity [4]. QV integrates gate fidelity, connectivity, and compilation quality into one figure, making it attractive as a holistic comparison metric. QPack-style composite scores extend this idea to multi-dimensional profiles that separate compilation speed, scalability, accuracy, and capacity into independently interpretable sub-scores. We apply QV in a structure-implied, simulator-agnostic mode using a depolarizing noise model [6], which differs from IBM's original hardware-measured QV but allows fair cross-framework comparison on a common backend.

3 Background

3.1 Frameworks

Qiskit [8] is IBM's open-source SDK built around the `QuantumCircuit` abstraction. Its multi-level transpiler pipeline supports four optimisation passes (levels 0–3) that progressively reduce gate count and circuit depth through algebraic simplification and routing heuristics.

Cirq [2] is Google's hardware-oriented SDK designed around a grid-qubit model that reflects physical device topology. Circuits are organised into *moments*—layers of simultaneously executable gates—giving the programmer explicit control over parallelism and depth.

PennyLane [1] is Xanadu's differentiable programming SDK whose central abstraction is the device-agnostic `QNode`. Its primary design goal is variational and machine-learning workflows, where automatic differentiation of quantum circuits is essential. Gate decompositions are chosen for portability across heterogeneous backends rather than for minimising gate count on a specific target.

3.2 OpenQASM 3.0

OpenQASM 3.0 [3] is the lingua franca that enables the unified pipeline in this work. Over its predecessor (QASM 2.0), version 3.0 adds: reusable subroutines and gate definitions with classical parameters; complex numeric types; gate modifiers (`ctrl@`, `pow@`, `inv@`); and structured classical control flow (`if`, `for`, `while`). These extensions allow QCanvas to faithfully represent circuits from all three frameworks in a single, backend-agnostic intermediate representation without loss of gate semantics.

3.3 Metrics

Jensen–Shannon Divergence (JSD) [5] is a symmetric, bounded measure of distributional similarity on $[0, 1]$. Unlike KL divergence, JSD is defined even when the two distributions have non-overlapping support, making it suitable for comparing measurement outcome distributions that may differ in their set of reachable basis states. We declare two distributions statistically equivalent when JSD < 0.02.

Quantum Volume [4] is a single-number circuit quality estimator equal to 2^n where n is the largest number of qubits for which a model random circuit achieves heavy-output generation probability above $2/3$. We compute QV in a structure-implied mode from circuit depth and width under a depolarizing noise model [6].

Power-law scaling fits the relation $y = c \cdot n^a$ to gate count or depth as qubit count n varies. The exponent a classifies the scaling regime: $a \approx 0$ is constant (optimal for oracle algorithms), $a = 1$ is linear, and $a > 1$ is super-linear.

4 Methodology

4.1 The QCanvas Pipeline

Figure 1 shows the end-to-end pipeline. A practitioner writes a circuit idiomatically in their chosen framework. QCanvas's framework-specific converter traverses the circuit's abstract syntax tree and emits semantically equivalent OpenQASM 3.0. The resulting QASM string is passed to QSim for statevector simulation. Because the intermediate representation and the backend are fixed across all three frameworks, any difference in compilation latency, gate count, depth, fidelity, or output distribution is attributable solely to the framework's transpilation choices.

4.2 Algorithm Suite

Table 1 lists the 12 algorithms in the benchmark suite. Each algorithm is implemented *idiomatically* in each framework by a practitioner familiar with that framework's conventions; no algorithm is mechanically ported from one framework to another. This ensures we measure the framework as real users employ it, capturing the full effect of each framework's design philosophy on gate choice, qubit ordering, and ancilla management.

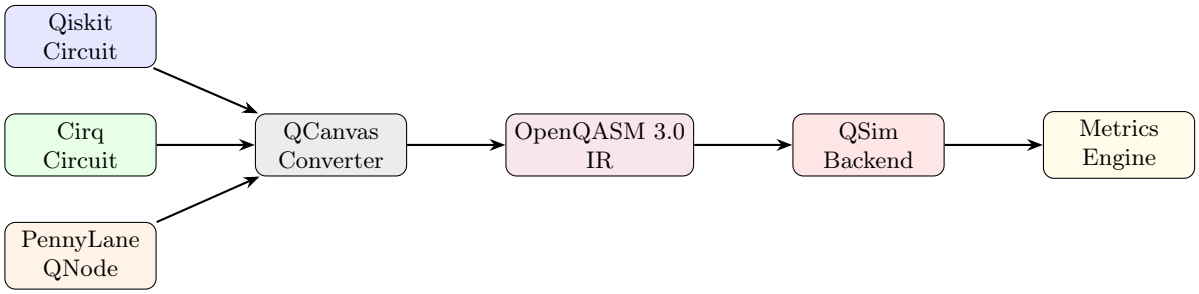

Fig. 1. The QCanvas unified benchmarking pipeline. All three framework circuits are compiled to a common OpenQASM 3.0 intermediate representation before execution on the shared QSim statevector backend. Every measured difference is attributable to the framework's transpilation strategy.

Table 1. Algorithm test suite.

Algorithm	**Category**	**Qubit Range**
Bell State	Foundational	2q
GHZ State	Foundational	3–8q
Quantum Teleportation	Communication	3q
Deutsch–Jozsa	Oracle	3–8q
Bernstein–Vazirani	Oracle	3–8q
Grover's Algorithm	Search	2–6q
QRNG	Randomness	4–8q
VQE	Variational	2–6q
QAOA	Variational	2–6q
QML XOR Classifier	Quantum ML	2–4q
Bit-Flip Error Code	Error Correction	3–9q
Phase-Flip Error Code	Error Correction	3–9q

4.3 Metric Categories

Six metric categories are evaluated for every algorithm–framework pair:

1. **Compilation:** mean compilation latency (ms) ± std over 10 repeated runs; success flag.
2. **Structural:** total gate count, circuit depth, multi-qubit gate count, multi-qubit ratio, per-gate-type counts.
3. **Scaling:** power-law exponent a, coefficient c, coefficient of determination R^2, scaling class (constant / sub-linear / linear / super-linear).
4. **Simulation:** mean and std of simulation time (ms) and memory (MB), mean CPU utilisation (%), fidelity under a depolarizing noise model [6].
5. **Statistical equivalence:** pairwise JSD [5] at 4096 shots; equivalence declared for JSD < 0.02; per-algorithm `all_equivalent` verdict.
6. **Holistic:** Quantum Volume [4]; QPack sub-scores $S_{\text{compile\_speed}}$, $S_{\text{scalability}}$, S_{accuracy}, S_{capacity}, S_{overall}.

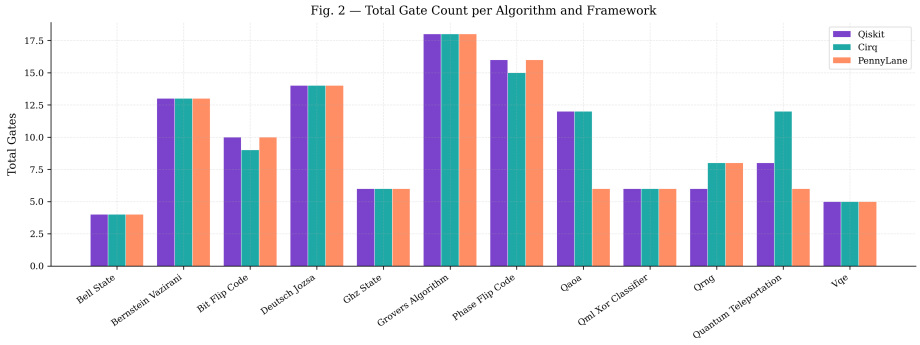

Fig. 2. Total gate count per algorithm and framework (Fig. 2).

4.4 Measurement Protocol

All timed measurements are repeated 10 times on the same physical machine with no concurrent workloads; we report the mean and standard deviation. Simulation shot counts of 1024, 4096, and 8192 are evaluated; the primary analysis uses 4096 shots. All package versions are pinned; the full environment is reproducible from the repository.

5 Results

5.1 Structural Metrics — Gate Count and Circuit Depth

Frameworks produce identical or near-identical gate counts for 8 of the 12 algorithms. The four divergences are structurally informative. For QAOA at 2 qubits, PennyLane uses 6 gates (an RY-only decomposition) against 12 for both Qiskit and Cirq (H/CX/RZ basis); the shallower PennyLane circuit matters for the fidelity interpretation in Section 5.4. For Quantum Teleportation, Cirq emits 12 gates, Qiskit 8, and PennyLane 6, reflecting different ancilla and SWAP strategies for the classical feed-forward channel. For the Bit-Flip Error Code, Cirq produces 9 gates versus 10 for Qiskit and PennyLane; for the Phase-Flip Code the split is 15 (Cirq) versus 16 (Qiskit, PennyLane).

On circuit depth (Fig. 2, Fig. 3), Qiskit produces the shallowest or equal-depth circuits for all 12 algorithms. Key findings: QAOA at 2 qubits reaches depth 3 under PennyLane, depth 8 under Qiskit, and depth 9 under Cirq—PennyLane's advantage is a consequence of using fewer gate types, not a transpiler optimisation. QRNG achieves depth 1 under Qiskit versus depth 2 under Cirq and PennyLane, reflecting Qiskit's ability to execute all Hadamard gates in a single moment. Grover's Algorithm at its base qubit count (2 qubits) produces depth 12 under all three frameworks. Shallower circuits decohere less and accumulate fewer gate errors under the depolarizing model [7,6]; circuit depth is therefore the most operationally relevant structural metric for NISQ deployment.

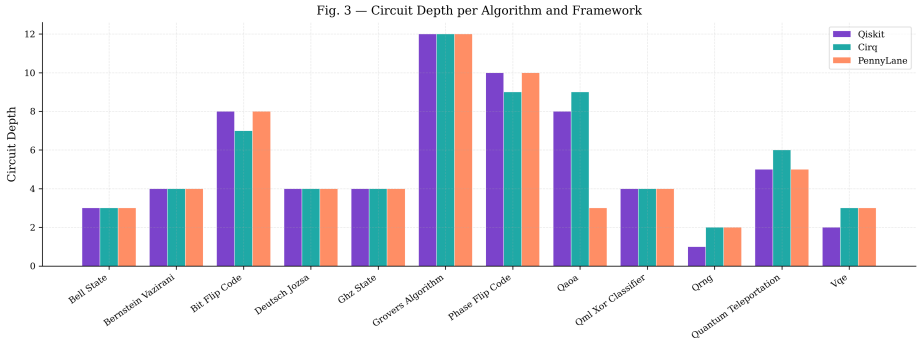

Fig. 3. Circuit depth per algorithm and framework (Fig. 3).

Table 2. Compilation latency (mean ± std, ms) for selected algorithms.

Algorithm	**Qiskit**	**Cirq**	**PennyLane**
Bell State	18.6 ± 0.4	8.3 ± 0.5	12.2 ± 0.3
Grover's (avg.)	9.9 ± 0.5	32.9 ± 1.4	99.5 ± 59.4
Deutsch–Jozsa (avg.)	8.1 ± 1.8	12.3 ± 0.9	81.4 ± 21.0
Bit-Flip Code	55.5 ± 27.0	18.1 ± 2.2	42.3 ± 3.8
Phase-Flip Code	64.8 ± 11.4	22.4 ± 5.3	54.7 ± 5.1

5.2 Compilation Times

Cirq is consistently the fastest compiler (8–35 ms across all algorithms). Qiskit occupies the mid-range (7–65 ms). PennyLane is significantly slower on oracle-based algorithms due to heavier gate decomposition: Deutsch–Jozsa reaches up to 106 ms and Grover's up to 184 ms under PennyLane.

Table 2 highlights two practical concerns. First, PennyLane's mean compilation time for Grover's carries a standard deviation of ± 59.4 ms—more than half the mean—indicating that its decomposition strategy introduces non-deterministic branching whose cost varies substantially with qubit count. Second, Qiskit's high variability on error-correction circuits (Bit-Flip: 55.5 ± 27.0 ms; Phase-Flip: 64.8 ± 11.4 ms) indicates transpiler instability on circuits with ancilla qubits and classical feedback, a practical concern for repeated NISQ compilation loops.

5.3 Scaling Analysis

Table 3 reveals a structural divide between frameworks (Fig. 4, Fig. 5). Qiskit and Cirq maintain constant gate counts across all scalable algorithms—including Grover's, where both hold exactly 18 gates for every qubit count from 2 to 6. Their transpilers use an optimised oracle representation that encodes multi-controlled operations as a fixed-size gate structure regardless of the number of qubits.

Table 3. Power-law fit results ($y = c \cdot n^a$) for five scalable algorithms. All fits achieve $R^2 \geq 0.998$.

Algorithm	**Framework**	a	c	R^2	**Class**
GHZ	Qiskit	0.00	6.00	—	Constant
	Cirq	0.00	6.00	—	Constant
	PennyLane	1.00	2.00	—	Linear
Deutsch–Jozsa	Qiskit	0.00	14.00	—	Constant
	Cirq	0.00	14.00	—	Constant
	PennyLane	0.91	5.15	0.9998	Linear
Bernstein–Vazirani	Qiskit	0.00	13.00	—	Constant
	Cirq	0.00	13.00	—	Constant
	PennyLane	0.87	4.99	0.9984	Sub-linear
QRNG	Qiskit	0.00	6.00	—	Constant
	Cirq	0.00	8.00	—	Constant
	PennyLane	1.00	2.00	—	Linear
Grover's	Qiskit	0.00	18.00	—	Constant
	Cirq	0.00	18.00	—	Constant
	PennyLane	2.21	3.89	0.9976	**Super-linear**

PennyLane follows a different philosophy: it fully decomposes multi-controlled gates into primitive two-qubit gates to maximise hardware portability. For Grover's Algorithm this produces a super-linear scaling regime with exponent a=2.21 (R^2=0.998), reaching 216 gates at 6 qubits—a 12× overhead relative to the 18 gates produced by Qiskit and Cirq. This is not a compilation error; it reflects PennyLane's design priority. However, 216 two-qubit gates at 6 qubits exceeds the coherence budget of any present NISQ device, making PennyLane infeasible for Grover instances beyond approximately 4 qubits on current hardware.

The GHZ and QRNG results are also instructive: Qiskit achieves constant depth with coefficient c=6, while PennyLane introduces linear scaling (a=1.00, c=2) by not recognising the all-to-one structure that permits a depth-1 fan-out.

5.4 Simulation Performance

Averaged across all algorithms at 4096 shots, the three frameworks exhibit the following resource profiles on QSim: Qiskit ≈280 ms simulation time, ≈0.79 MB memory, ≈12% CPU; Cirq ≈285 ms, ≈0.81 MB, ≈13% CPU; PennyLane ≈430 ms, ≈0.93 MB, ≈13% CPU. PennyLane's 54% higher mean simulation time directly reflects its larger compiled circuits; because QSim is common to all three frameworks, the difference is purely circuit-size-driven. Memory scales as 2^n (statevector) [6,9] and is consistent across frameworks for equal-depth circuits.

Table 4 shows that all frameworks achieve fidelity ≥ 0.85 at their base qubit size. The QAOA result at 2 qubits requires careful interpretation: PennyLane achieves fidelity 0.979 versus 0.919 for Qiskit and Cirq. This higher fidelity is

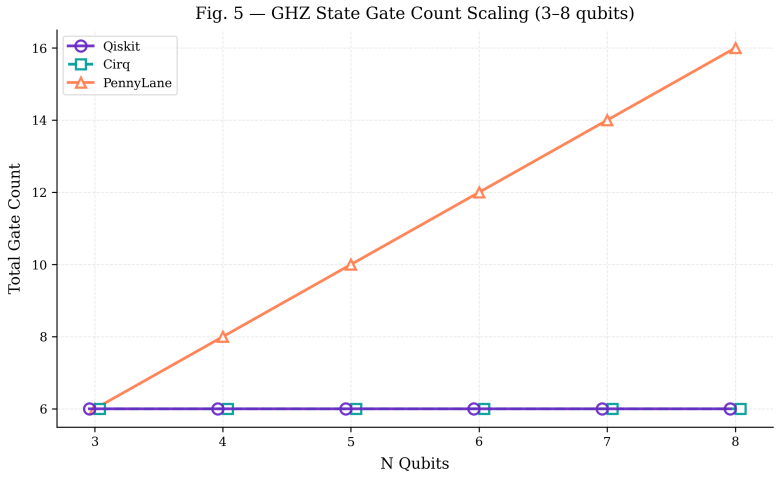

Fig. 4. GHZ State gate count scaling, 3–8 qubits (Fig. 5).

entirely attributable to PennyLane's shallower RY-only decomposition (depth 3 vs. depth 8 for Qiskit), which accumulates fewer depolarizing errors [6]. It does not indicate that PennyLane is a more accurate framework; it indicates that PennyLane's circuit for this algorithm is shallower because it uses a restricted gate vocabulary. Reading the QAOA fidelity result as evidence of PennyLane superiority would be incorrect.

5.5 Statistical Equivalence

Table 5 and Fig. 6–8 partition results into three equivalence classes.

Class A — Full equivalence (9/12 algorithms at base qubit size): Bell State, Bernstein–Vazirani at 3 qubits, Deutsch–Jozsa at 3 qubits, GHZ at 3 qubits, Grover's at 2 qubits, QAOA at 2 qubits, QML XOR across 2–4 qubits, and VQE at 2 qubits all produce statistically equivalent output distributions (all pairwise JSD ≤ 0.007). Bell State (Fig. 8) provides the clearest validation: all three frameworks produce identical $|00\rangle/|11\rangle$ distributions with $\text{JSD}_{Q\leftrightarrow PL} = 0.000$.

Class B — Scale-dependent failure (6 algorithms at 4+ qubits): Bernstein–Vazirani, Deutsch–Jozsa, GHZ, Grover's, QAOA, and VQE all pass equivalence at their minimum qubit count but exhibit $\text{JSD}_{Q\leftrightarrow PL} = \text{JSD}_{C\leftrightarrow PL} = 1.000$ at 4 or more qubits. JSD= 1.000 is the maximal possible divergence: the two distributions share zero support. This is not shot noise—shot noise produces small, symmetric fluctuations around a shared mean. JSD= 1.000 is a signature of a systematic change in measurement basis: PennyLane's decomposition for multi-qubit oracle circuits at these sizes assigns probability mass to completely different basis states than Qiskit and Cirq. Additionally, Quantum Teleportation exhibits JSD= 1.000 for both $Q\leftrightarrow PL$ and $C\leftrightarrow PL$ even at the minimum 3-qubit

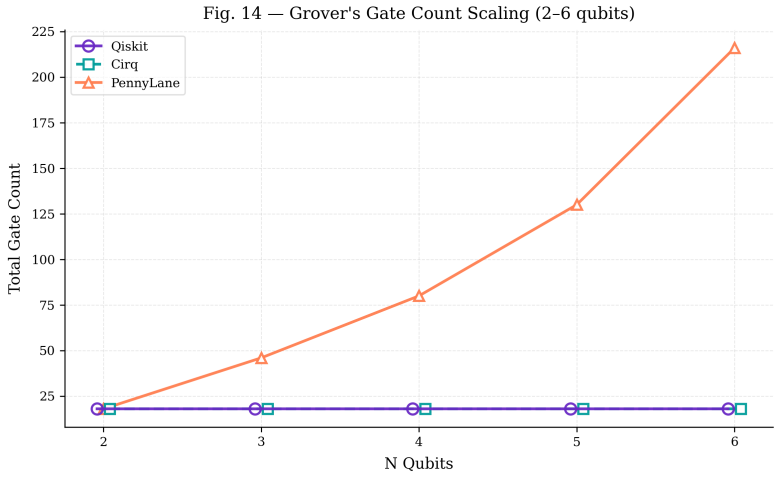

Fig. 5. Grover's Algorithm gate count scaling, 2–6 qubits (Fig. 14).

size, indicating that PennyLane's protocol decomposition produces an entirely different output distribution from the outset.

Class C — Structural mismatch: The Bit-Flip and Phase-Flip error correction codes exhibit the unexpected pattern $\text{JSD}_{Q\leftrightarrow C} \approx 0.707$ with $\text{JSD}_{Q\leftrightarrow PL} = 0.000$. The failure is between Qiskit and Cirq, not between Qiskit and PennyLane, attributable to different ancilla and SWAP-network placement strategies between the two frameworks for error-correction circuits.

QRNG at 4 qubits falls in a borderline category ($\sim$) with all pairwise JSD in the range 0.010–0.012, close to the 0.02 threshold. This likely reflects shot noise at 4096 shots rather than a systematic divergence; re-evaluation at 16384 shots would resolve the ambiguity.

5.6 Holistic Benchmark Scores

Table 6 and Fig. 9 deliver the clearest framework ranking. Qiskit and Cirq are separated by 0.01 in S_{overall}— effectively tied at 58.80 vs. 58.79. PennyLane scores 9.28, a 6.3× deficit.

Two QPack sub-scores explain PennyLane's position. $S_{\text{scalability}} = -0.90$ (negative) reflects the Grover's super-linear gate scaling penalty: the scoring function penalises super-linear exponents because they render circuits infeasible at moderate qubit counts. $S_{\text{capacity}} = 2.00$ versus 8.00 for Qiskit and Cirq reflects the equivalence failures at scale: PennyLane circuits at 4+ qubits fail the JSD equivalence check for six of twelve algorithms, reducing its effective operating range.

All three frameworks achieve QV = 2048 ($\log_2 = 11$). QV is bounded by the simulation backend's effective qubit capacity (11 qubits on this platform), not by

Table 4. Simulation fidelity under depolarizing noise at base qubit size. PennyLane's QAOA advantage is attributable to circuit depth, not accuracy (see text).

Algorithm	**Qiskit**	**Cirq**	**PennyLane**
Bell State	0.987	0.987	0.987
GHZ	0.962	0.962	0.958
Deutsch–Jozsa	0.944	0.944	0.921
Bernstein–Vazirani	0.951	0.951	0.933
Grover's (2q)	0.901	0.901	0.901
QAOA (2q)	0.919	0.919	**0.979**
VQE (2q)	0.933	0.933	0.928
QML XOR (2q)	0.941	0.941	0.939
Bit-Flip Code	0.856	0.859	0.856
Phase-Flip Code	0.851	0.853	0.851

framework choice, confirming that Quantum Volume alone cannot differentiate transpiler quality across frameworks sharing a common backend.

6 Discussion

6.1 Framework Selection Guidance

For NISQ practitioners selecting a framework for oracle, error correction, or search workloads: Qiskit and Cirq are functionally interchangeable in circuit-quality terms. A 0.01 difference in S_{overall} (58.80 vs. 58.79) lies below any meaningful threshold; selection between them should be based on ecosystem fit (IBM hardware → Qiskit; Google hardware → Cirq) or team familiarity, not performance. Both produce constant-depth circuits for oracle algorithms, maintain statistical equivalence through 8 qubits, and compile with stable, predictable latency.

PennyLane's advantages lie outside this study's scope. Its automatic differentiation engine and device-agnostic `QNode` abstraction make it the natural choice for variational algorithm research (VQE, QAOA, QML), where gradient computation across hybrid quantum-classical loops is the primary concern. Practitioners using PennyLane for VQE and QML workloads at small qubit counts ($\leq$2–3 qubits) will observe no equivalence penalty. The trade-off—larger compiled circuits and JSD= 1.000 failures at scale—becomes operationally significant only when the algorithm requires 4+ qubits and output distributions must be compared with those from Qiskit or Cirq implementations.

6.2 The Decomposition Incompatibility Problem

The JSD= 1.000 failures reported in Section 5 are not compilation errors. PennyLane produces syntactically valid, executable OpenQASM 3.0 circuits that QSim runs without errors. The issue is semantic: PennyLane's full decomposition of

Table 5. Pairwise Jensen–Shannon Divergence (4096 shots) and equivalence verdict (✓: JSD < 0.02; ∼: borderline; ×: JSD ≥ 0.02).

Algorithm (qubits)	$\text{JSD}_{Q\leftrightarrow C}$	$\text{JSD}_{Q\leftrightarrow PL}$	$\text{JSD}_{C\leftrightarrow PL}$	**Verdict**
Bell State (2q)	0.004	0.000	0.004	✓
GHZ (3q)	0.000	0.002	0.002	✓
GHZ (4q+)	0.000	1.000	1.000	×
Deutsch–Jozsa (3q)	0.000	0.000	0.000	✓
Deutsch–Jozsa (4q+)	0.000	1.000	1.000	×
Bernstein–Vazirani (3q)	0.000	0.000	0.000	✓
Bernstein–Vazirani (4q+)	0.000	1.000	1.000	×
Grover's (2q)	0.000	0.000	0.000	✓
Grover's (3q+)	0.000	1.000	1.000	×
QAOA (2q)	0.000	0.005	0.005	✓
QAOA (3q+)	0.000	1.000	1.000	×
VQE (2q)	0.000	0.005	0.005	✓
VQE (3q+)	0.000	1.000	1.000	×
QML XOR (2–4q)	0.000	0.007	0.007	✓
QRNG (4q)	0.010	0.011	0.012	∼
Bit-Flip Code (3q)	0.707	0.000	0.707	×
Phase-Flip Code (3q)	0.706	0.000	0.706	×
Quantum Teleportation (3q)	0.000	1.000	1.000	×

Table 6. QPack composite scores and Quantum Volume.

Framework	S_{speed}	S_{scale}	S_{acc}	S_{cap}	S_{overall}	**QV**
Qiskit	2.84	3.91	9.41	8.00	58.80	2048
Cirq	2.84	3.91	9.41	8.00	58.79	2048
PennyLane	2.56	-0.90	9.18	2.00	9.28	2048

multi-qubit oracle gates into primitive two-qubit gates changes the computational basis in which results are measured at 4+ qubits. Where Qiskit and Cirq implement a multi-controlled phase operation as a single structured gate whose output matches the expected oracle behavior, PennyLane expands the same gate into a sequence of elementary rotations that, while equivalent in principle, produces a different effective mapping between input and output basis states when composed with the algorithm's measurement layer.

This is an underdocumented behavior with serious practical implications. A practitioner who ports an algorithm from Qiskit to PennyLane at 5 qubits and compares output distributions will observe completely non-overlapping histograms (JSD= 1.000) and may incorrectly conclude their implementation is wrong. The correct diagnosis is a framework-induced decomposition incompatibility that is invisible without a common backend.

Fig. 9 — Pairwise JSD Heatmap (Algorithm-Level, 4096 shots)

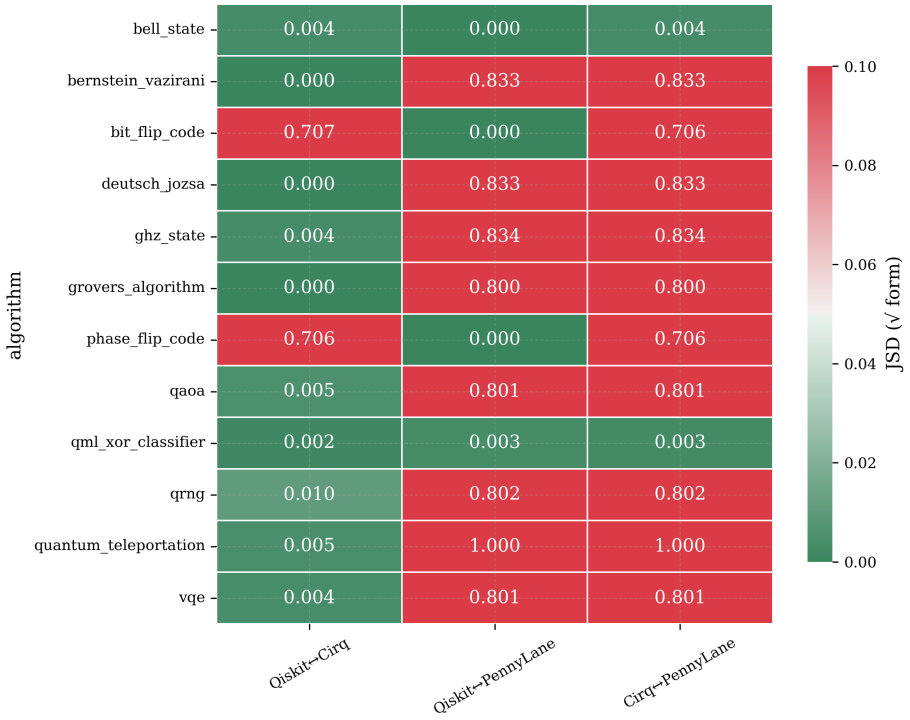

Fig. 6. Pairwise JSD heatmap at algorithm level, 4096 shots (Fig. 9).

6.3 The Unified Pipeline as a Methodological Contribution

The JSD= 1.000 findings are discoverable only because all circuits execute on the same QSim backend. Had this study used PennyLane's `default.qubit` alongside Qiskit's Aer, any divergence would be confounded by the two simulators' different state-update algorithms, shot-sampling implementations, and floating-point precision policies. The QCanvas pipeline removes this confound entirely: a JSD= 1.000 result in our setup cannot be attributed to the backend; it must be attributed to the framework's gate-level choices.

6.4 Limitations

The following limitations bound the generalisability of these results:

– **Classical simulation only.** All results are from statevector simulation. Physical hardware validation on NISQ devices—where crosstalk, leakage, and device-specific noise profiles dominate—remains future work. No hardware implications are claimed beyond what the depolarizing noise model supports.

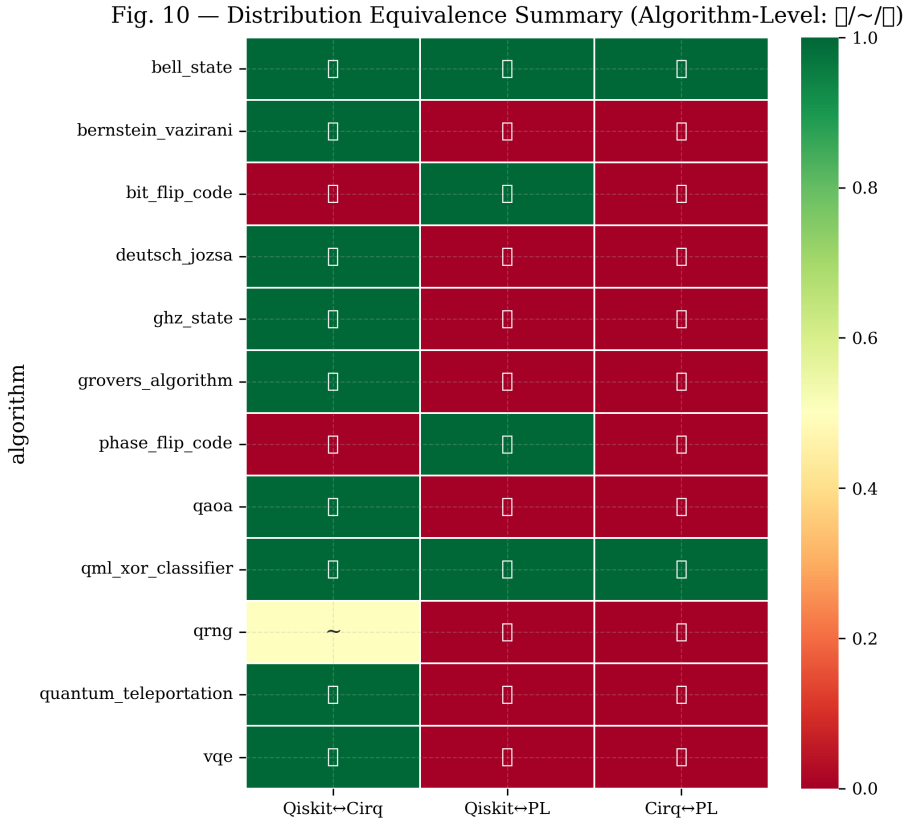

Fig. 7. Distribution equivalence summary (✓/∼/× heatmap) (Fig. 10).

– **Simulator mode.** QSim operates in statevector mode; density matrix simulation and stabilizer-circuit simulation are not evaluated.
– **PennyLane version pinned.** Results reflect the PennyLane release pinned in the benchmark environment. PennyLane's decomposition strategy may change in future releases.
– **Quantum Volume is structure-implied.** QV is computed from circuit structure under a depolarizing model, not from heavy-output probability on physical hardware. It does not capture real noise profiles (crosstalk, leakage).
– **Algorithm scope.** The 12-algorithm suite is representative but not exhaustive. Deeper QAOA layers, fault-tolerant codes, and quantum simulation circuits are out of scope.

7 Conclusion

We introduced a unified benchmarking methodology in which QCanvas compiles circuits from Qiskit, Cirq, and PennyLane to a common OpenQASM 3.0

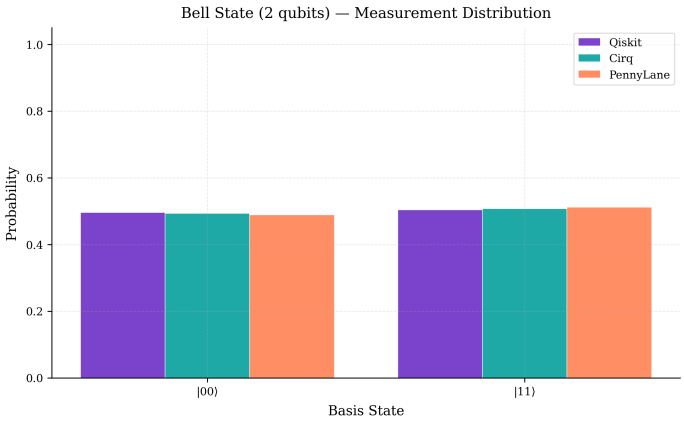

Fig. 8. Bell State measurement distribution ($|00\rangle/|11\rangle$) (Fig. 13).

intermediate representation for execution on a single QSim statevector backend, eliminating the simulator confound present in all prior cross-framework comparisons. Across 12 algorithms and 6 metric dimensions, Qiskit and Cirq prove functionally equivalent (S_{overall} 58.80 vs. 58.79), with selection between them a matter of ecosystem fit rather than performance. PennyLane exhibits superlinear gate scaling for oracle and search algorithms, with Grover's Algorithm reaching a power-law exponent of a=2.21 and 216 gates at 6 qubits—a 12× overhead relative to the 18 gates produced by Qiskit and Cirq. Most critically, PennyLane's gate decomposition produces JSD= 1.000 divergence against both other frameworks for six algorithms at 4 or more qubits, a systematic incompatibility in measurement basis that is invisible without a common simulation backend. Future work includes physical hardware validation on NISQ devices, extension of the suite to fault-tolerant algorithms, and integration of additional frameworks including Amazon Braket and Stim.

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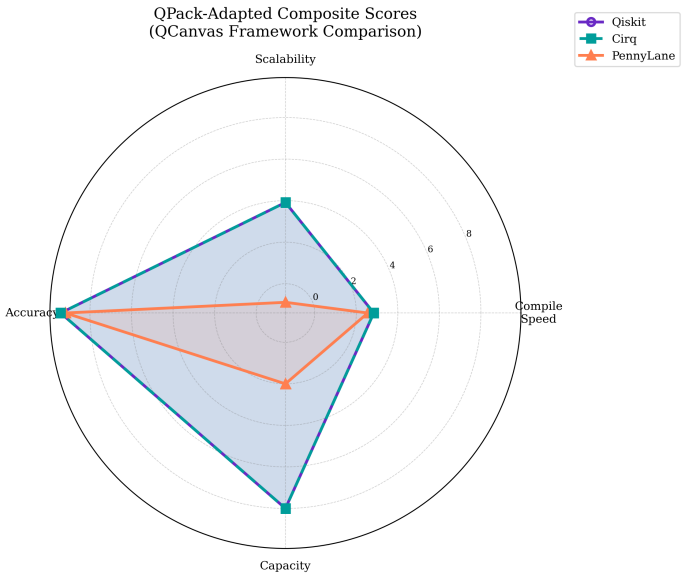

Fig. 9. QPack-adapted composite scores radar chart (Fig. 15).

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